

Electromagnetic compatibility — Product family standard for lifts, escalators and moving walks — Emission

London Underground Ltd

The European Standard EN 12015:2004 has the status of a
British Standard

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National foreword

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The UK participation in its preparation was entrusted to Technical Committee MHE/4, Lifts, hoists and stairlifts, which has the responsibility to:

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Summary of pages

This document comprises a front cover, an inside front cover, the EN title page, pages 2 to 16, an inside back cover and a back cover.

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Electromagnetic compatibility - Product family standard for lifts, escalators and moving walks - Emission

Compatibilité électromagnétique - Norme famille de
produits pour ascenseurs, escaliers mécaniques et trottoirs
roulants -Emission

Elektromagnetische Verträglichkeit - Produktfamilien-Norm
für Aufzüge, Fahrtreppen und Fahrsteige - Störaussendung

This European Standard was approved by CEN on 1 April 2004.

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Foreword

This document (EN 12015:2004) has been prepared by Technical Committee CEN/TC 10 “Lifts, escalators and moving walks”, the secretariat of which is held by AFNOR.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 2005, and conflicting national standards shall be withdrawn at the latest by June 2006.

This document supersedes EN 12015:1998.

This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA, which is an integral part of this document.

The limits given in this standard recognize the fact that the product family covers a total range of lifts, escalators and moving walks used in residential buildings, offices, hospitals, hotels, industrial plants etc. and that lifts, escalators and moving walks are deemed to have their own dedicated power supply and be connected with the consent of the supply authority to a low impedance source.

The related EMC product family standard for immunity is:

EN 12016, *Electromagnetic compatibility — Product family standard for lifts, escalators and passenger conveyors — Immunity*.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

Introduction

This European Standard has been prepared to provide one means of complying with the requirements of the Electromagnetic Compatibility (EMC) Directive. The requirements of this European Standard have been specified so as to ensure a level of electromagnetic emission which will cause minimal disturbance to other equipment. The levels, however, do not cover the following cases:

- where the probability of an occurrence likely to produce emissions in excess of those which would normally be experienced is extremely low, e.g. the emergency stopping of a lift, escalator or moving walk under a fault condition;
- where highly susceptible apparatus will be used in the close proximity of the equipment covered by this standard, in which case further measures may have to be taken to:
 - reduce the level of electromagnetic emission to below that specified in this standard; or
 - increase the immunity of the affected apparatus.

The given emission limits, are on the basis that equipment of the product family range may be installed both indoor and outdoor in all types of building, involves the switching of heavy currents and high inductive loads and, generally, is connected to a low voltage system.

Due to the size of an installed lift, it becomes impracticable to test the total assembly either in a test laboratory or in situ where the uncontrolled environment may also influence the test procedures and results. This applies also to measurements within the car. Similar considerations regarding dimensions apply equally to the testing of escalators and moving walks.

The following explains the rational to the revision of the standard EN 12015:1998.

a) Important changes

Introduced requirements to control the emissions below 30 MHz of the drive to machine/motor connection. The emission limits are independent of the magnitude of the conducted current. Limits and test method are referred to EN 55014-1. Regarding other ports, the radiated tests above 30 MHz cover the cable connections and there are no known problems below 30 MHz.

Introduced requirements to control mains electricity supply harmonic emissions and voltage fluctuations.

NOTE The radiation measurements in Table 1 have been harmonised with EN 55011.

The term "installation" has been changed to "system". This is due to the fact that official interpretation defines that fixed installations are not covered by the conformity assessment requirements of the EMC Directive, valid for apparatus and systems. The scope of the standard is applicable to the apparatus and assembly of apparatus of lifts and escalators and assembly into systems.

b) Environmental issues

Lifts, escalators and moving walks are systems whose apparatus and assembly of apparatus are distributed (and some of which move) throughout the building. The definition in EMC terms of the use of the building (residential or industrial) cannot be predetermined or assumed to be fixed. Therefore, to cover requirements in all cases, no differentiation between environments has been made and a single set of limits has been maintained. This set of high frequency limits is based on the industrial limits of EN 61000-6-4 and is known to be above the usual limits for the residential environment. This is justified by the experience that systems in compliance with EN 12015:1998 have not been known to cause EMC interference with regard to mains and radiated emissions above 30 MHz.

Regarding conducted limits, these are also based on the fact that the supply cabling is separated from other building supplies at least up to the point of common coupling (PCC). Additionally, system wiring is segregated in accordance with the manufacturers' specifications.

Regarding radiated emission limits above 30 MHz, the lift, escalator or moving walk area is separated, to a large extent, from domestic appliances.

All the limits used take into account that systems have to comply with the safety protection requirements regarding earth leakage currents. The application of more stringent limits than have been shown to be adequate, would require the use of larger filters (both inductance and capacitance). The use of these will increase the susceptibility of the system to low electricity mains supply conditions and increase earth leakage currents. There is also an increase in power dissipation in the filter, causing (in the general case) increased energy consumption, additional ventilation and/or cooling requirement for the building. This is especially valid for higher duty systems.

1 Scope

This European Standard specifies the emission limits in relation to electromagnetic disturbances and test conditions for lifts, escalators and moving walks, which are intended to be permanently installed in buildings. These limits however, may not provide full protection against disturbances caused to radio and TV reception when such equipment is used within distances given in Table 1.

This standard addresses the environmental conditions stated in the EN 81 series of standards and EN 115 (humidity, temperature, etc.), so far as they are related to EMC performance.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 81, *Safety rules for the construction and installation of lifts.*

EN 115, *Safety rules for the construction and installation of escalators and passenger conveyors.*

EN 55011, *Industrial, scientific and medical (ISM) radio-frequency equipment - Radio disturbance characteristics - Limits and methods of measurement (CISPR 11:1997, modified).*

EN 55014-1:2000, *Electromagnetic compatibility - Requirements for household appliances, electric tools and similar apparatus - Part 1: Emission (CISPR 14-1:2000).*

EN 61000-3-11, *Electromagnetic compatibility (EMC) — Part 3-11: Limits — Limitation of voltage changes, voltage fluctuations and flicker in low-voltage supply systems for equipment with rated current up to 75A and subject to conditional connection (IEC 61000-3-11:2000).*

EN 61000-6-3, *Electromagnetic compatibility (EMC) — Part 6-3: Generic standards — Emission standard for residential, commercial and light-industrial environments (IEC 61000-6-3:1996, modified).*

EN 61000-6-4, *Electromagnetic compatibility (EMC) — Part 6-4: Generic standards — Emission standard for industrial environments (IEC 61000-6-3:1997, modified).*

IEC 60050-161, *International Electrotechnical Vocabulary; chapter 161: Electromagnetic compatibility.*

IEC/TR2 61000-3-4:1998, *Electromagnetic compatibility (EMC) - Part 3-4: Limits - Limitation of emission of harmonic currents in low-voltage power supply systems for equipment with rated current greater than 16 A.*

3 Terms and definitions

For the purposes of this European Standard, the terms and definitions given in EN 61000-6-3, EN 61000-6-4 and IEC 60050-161 and the following apply.

3.1 system

lift, escalator or moving walk comprising assembly of apparatus with electrical and electronic equipment and interconnections

NOTE See Figure 1 and Figure 2 as examples.

3.2**assembly of apparatus**

arrangement of interconnected apparatus, which can be tested together

NOTE See Figure 1 and Figure 2 as examples.

3.3**apparatus**

assembly of components with an intrinsic function as defined by its manufacturer

NOTE 1 See Figure 1 and Figure 2 as examples.

NOTE 2 Safety components defined by annex IV of the Lifts Directive are considered as apparatus.

3.4**Root Mean Square (rms)**

effective value of the current

3.5**Total Harmonic Distortion (THD)**

ratio of the rms value of the harmonics to the rms value of the fundamental

NOTE THD is calculated from the equation below. I_n means the rms value of the current of the n^{th} harmonic and I_1 means the rms value of the fundamental current:

$$THD = \sqrt{\sum_{n=2}^{40} \left(\frac{I_n}{I_1} \right)^2}$$

3.6**Partial Weighted Harmonic Distortion (PWHD)**

ratio of the rms value of a selected group of higher order harmonics (here beginning from the 14th harmonic), weighted with the harmonic order n , to the rms value of the fundamental

NOTE PWHD is calculated from the equation below. I_n means the rms value of the current of the n^{th} harmonic and I_1 means the rms value of the fundamental current:

$$PWHD = \sqrt{\sum_{n=14}^{40} n \left(\frac{I_n}{I_1} \right)^2}$$

3.7**balanced three phase system**

system connected to the three phases and designed in such a way that, for rated conditions, the rms value of the current in each of the three phases differs by not more than 20 %.

NOTE The neutral conductor is not used as a current carrying conductor under normal operating conditions.

3.8**Point of Common Coupling (PCC)**

point in the public mains network which is nearest to the system and to which, other equipment may be connected

3.9**short circuit power (S_{sc})**

value of the three phase short circuit power calculated from nominal voltage of the system U_n and impedance Z at PCC:

$$S_{sc} = \frac{U_n^2}{Z}$$

3.10

rated apparent power (S_{equ})

value calculated from the phase to phase voltage U_i and the rated rms line current I_{equ} of the apparatus/assembly of apparatus, using the equation:

$$S_{\text{equ}} = \sqrt{3} \cdot U_i \cdot I_{\text{equ}}$$

3.11

short circuit ratio (R_{sce})

ratio of the short circuit power of the source to the apparent power of the load(s).

NOTE For apparatus/assembly of apparatus connected to a three phase electricity supply it is equal to:

$$R_{\text{sce}} = \frac{S_{\text{sc}}}{S_{\text{equ}}}$$

3.12

port

particular interface of the specified apparatus/assembly of apparatus with the external electromagnetic environment

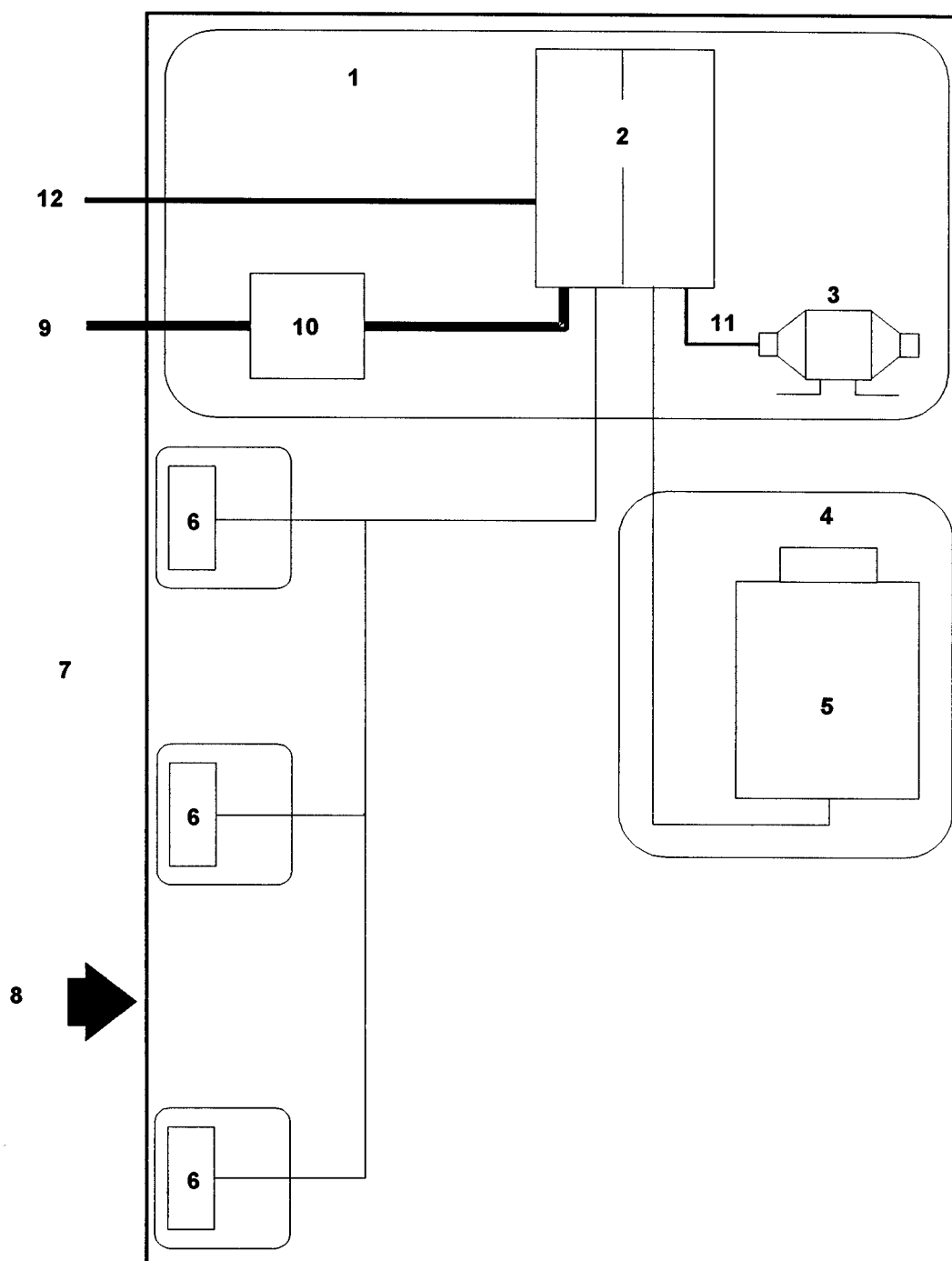
NOTE See Figure 3 as example.

3.13

enclosure port

boundary of the apparatus/assembly of apparatus through which electromagnetic fields can radiate or impinge

NOTE See Figure 3 as example.

**Key**

Assembly of apparatus

1 Machinery space

2 Main control/control cabinet

3 Machine

4 Door control

5 Lift car

6 Apparatus installed at the landing (e.g. push buttons, indicators)

7 Landings

8 System boundary

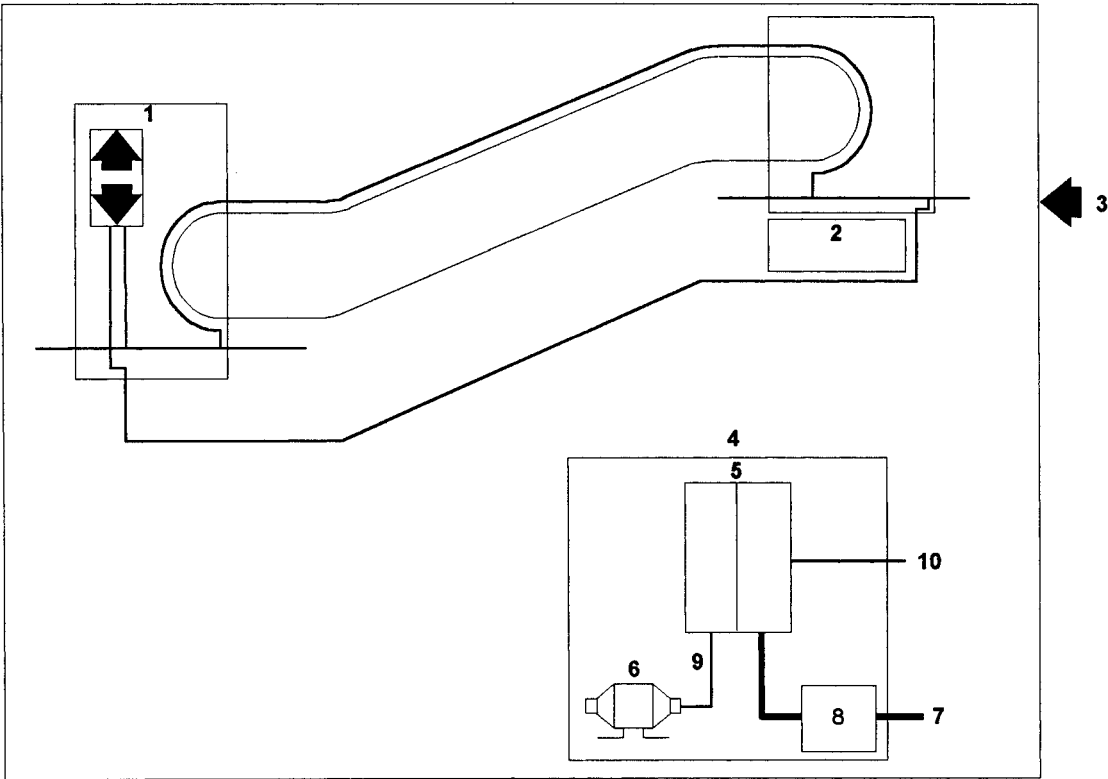
9 AC – and/or DC power ports

10 Main switch

11 Output power port

12 Signal and control ports

Figure 1 — EMC model example (emission) for lift systems



Key



- | | | | |
|------|---|----|----------------------------|
| 1 | Control panels | 6 | Machine |
| 2 | Machinery space (see 4 to 10) | 7 | AC – and/or DC power ports |
| NOTE | The machinery space can also be an external room. | | |
| 3 | System boundary | 8 | Main switch |
| 4 | Machinery space | 9 | Output power port |
| 5 | Main control/control cabinet | 10 | Signal and control ports |

Figure 2 — EMC model example (emission) for escalator and moving walk systems

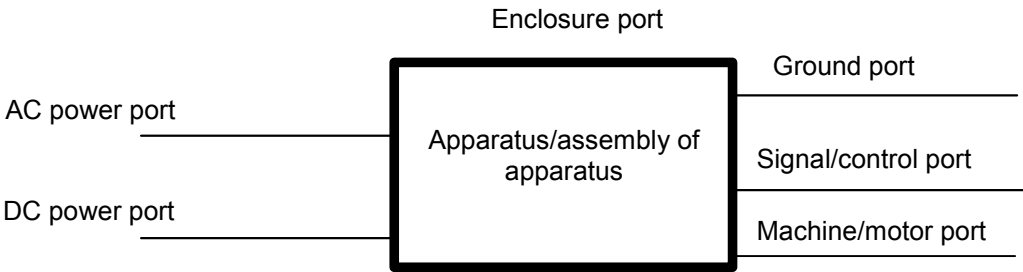


Figure 3 — Examples of ports

4 Test set-up procedure

4.1 General

4.1.1 The measurements shall be made in the operation mode producing the highest emissions consistent with normal applications. An attempt shall be made to maximise the emissions by varying the position of the test sample in the test facility.

4.1.2 It is not always possible to measure emission for every function of the apparatus or assembly of apparatus. In such cases the most critical period of operation shall be selected under normal operating modes.

4.1.3 If the normative standards do not specify any other conditions then the tests shall be carried out at a single set of environmental conditions within the manufacturers specified operating range of temperature, humidity, pressure and supply voltage.

4.1.4 Measurements shall be taken in well-defined and reproducible conditions for each test. The configuration and mode of operation during measurement shall be precisely recorded.

4.2 Radiated and conducted radio frequency emissions

4.2.1 The tests, test methods, characteristics of the tests and test set-ups shall be as stated in EN 55011.

4.2.2 Travelling cables or any other cables likely to be more than 5 m long shall be represented by a sample of at least 5 m long connected to the relevant port for the purpose of testing for radiative emission.

4.2.3 If the apparatus has a large number of similar ports or ports with many similar connections, then a sufficient number shall be selected to simulate actual operating conditions and to ensure that all the different types of termination are covered.

4.2.4 Measurement shall be taken at the enclosure ports (radiative), A.C. mains and output power ports (conductive) of the apparatus/assembly of apparatus.

4.3 Mains supply harmonics

The tests, test methods, characteristics of the tests and test set-ups shall be as stated in IEC/TR2 61000-3-4.

As the supply authorities do not guarantee a defined mains impedance, the mains short circuit power may vary from location to location. As these parameters are normally not known in advance, a practical assumption has to be made, based on values already known from other lifts/escalators and moving walks already in service.

Due to that fact, a short circuit ratio $R_{scc} = 250$ is defined as an average value for lifts, escalators and moving walks. As these are normally within the limits of the definition for a balanced three phase system, the values given in Table 4 are taken from the standards related to these kinds of systems.

5 Applicability of tests

The application of tests for evaluation of levels of emission depends on the type of apparatus/assembly of apparatus, its configuration, ports, technology and operating conditions.

5.1 It might be determined from consideration of the electrical characteristics and usage of a particular apparatus/assembly of apparatus, that some of the tests are inappropriate and therefore unnecessary. In such a case the decision and justification not to test shall be recorded.

5.2 Where deviations from the test methods specified in 4.2.1 and 4.3, are applied, such deviations shall be justified and recorded.

6 Emission limits

6.1 Enclosure ports (radiative)

The electromagnetic emission levels measured at each enclosure port (radiative) of the apparatus/assembly of apparatus shall not exceed the limits specified in Table 1. These limits are not applicable to in-situ measurements.

6.2 AC mains ports (conductive)

6.2.1 The electromagnetic emission levels measured at each A.C. mains port (conductive) of the apparatus / assembly of apparatus operating at less than 1 000 V rms shall not exceed the limits specified in Table 2. Different limits apply to emissions resulting from certain rates of impulse noise, as stated in 6.4.

6.2.2 The harmonics emission levels measured at each A.C. mains port (conductive) of the apparatus/assembly of apparatus, operating at less than or equal to 600 V rms shall not exceed the limits specified in Table 4. Different limits apply to emission resulting from certain operating conditions, also defined in this table.

6.3 Output power ports (conductive)

The electromagnetic emission levels at each machine/motor port (conductive) of the apparatus/assembly of apparatus shall not exceed the limits specified in Table 3. If shielded connectors and shielded cables, in accordance with the apparatus/assembly of apparatus manufacturer's specification are used for machine/motor ports, or the cables are less than or equal to 2 m long, then measurements at those ports are not necessary.

6.4 Impulse noise

Electromagnetic emission levels resulting from impulse noise (clicks), measured as for 6.2.1, shall not exceed the limits specified in Table 2 if the clicks occur more frequently than 30 times per minute. Electromagnetic emission levels resulting from clicks which occur between 0,2 times and 30 times per minute shall not exceed the limits specified in Table 2 raised by a value of:

$$20 \log_{10} \frac{30}{N} \text{ dB}(\mu\text{V})$$

(where N is the number of clicks per minute). These limits do not apply for the exceptions given in EN 55014-1:2000, 4.2.3.

6.5 Voltage fluctuations

The requirements of EN 61000-3-11 are applicable.

Voltage fluctuations are dependent on the impedance of the mains supply for an individual system and the characteristics of the apparatus/assembly of apparatus. The manufacturer shall document the maximum impedance of the electricity mains supply for the system.

NOTE Apparatus/assembly of apparatus using controlled speed drives are known not to be a cause of voltage flicker disturbance. However, direct on line or star-delta starting of escalator motors, hoisting and hydraulic pump motors and repetitive direct switching of highly inductive loads (e.g. transformer) should be treated with caution.

6.6 Mains supply current harmonics

The limits of Table 4 apply.

6.7 Measurement

6.7.1 Radiated and conducted radio frequency emission

To determine conformity to the requirements of 6.1 and 6.2.1, emission levels shall be measured by the methods specified in EN 55011 and under the conditions specified in clause 4 of this standard.

To determine conformity to the requirements of 6.3, the measurement method as defined in EN 55014-1 for load terminals shall be used.

6.7.2 Mains Supply Harmonics

If the ratios I_n/I_1 for further evaluation are determined by measurements, they shall be derived from instantaneous values of the fundamental and harmonic currents.

The ratios I_n/I_1 used for comparison with the limits given in Table 4 shall be based on a fundamental current that is at least the rated current of the system defined by the manufacturer.

Table 1 — Emission limits for enclosure (radiative) ports

Frequency range MHz	Limits for measurements on a test site at 10 m distance ^a dB(μV/m)
$30 \leq F < 230$	40 quasi peak
$230 \leq F \leq 1000$	47 quasi peak
^a These limits are based on those specified in EN 61000-6-4. Measurements made at a distance of less than 10 m shall be made in accordance with EN 55011. Measurements shall not be taken at a distance of less than 3 m.	

Table 2 — Emission limits for A.C. mains (conductive) ports

Frequency range MHz	Limits dB(μV)		
	Measurement with rated mains input current ^a		
	< 25 A	25-100 A	> 100 A ^c
$0,15 \leq F < 0,5$	79 quasi peak 66 average	100 quasi peak 90 average	130 quasi peak 120 average
$0,5 \leq F < 5,0$	73 quasi peak 60 average	86 quasi peak 76 average	125 quasi peak 115 average
$5,0 \leq F < 30$	73 quasi peak 60 average	90 to 70 ^b quasi peak 80 to 60 ^b average	115 quasi peak 105 average
^a The current for which the apparatus has been designed. ^b Decreasing with logarithm of frequency. ^c This assumes a dedicated power supply from a specific transformer.			

Table 3 — Emission limits for output power (conductive) ports

Frequency range MHz	Limits dB(μV)
	Measurement with rated output current
$0,15 \leq F < 0,5$	80 quasi peak 70 average
$0,5 \leq F < 5,0$	74 quasi peak 64 average
$5,0 \leq F < 30$	74 quasi peak 64 average

Table 4 — Harmonic distortion

Environmental Phenomena	Test set-up	Units	Limits for permanent emission ^a	Limits for short duration ^b
Harmonic distortion	IEC/TR2 61000-3-4	I_5/I_1 [%]	30	45
		I_7/I_1 [%]	18	27
		I_{11}/I_1 [%]	13	19
		I_{13}/I_1 [%]	8	12
Harmonic distortion factor	IEC/TR2 61000-3-4	THD [%]	35	52
		PWHD [%]	39	58
NOTE 1 The relative value of even harmonics should not exceed 16/n [%].				
NOTE 2 The relative value of odd triplen harmonics should not exceed 9/n [%].				
^a The given limits are based on IEC/TR2 61000-3-4:1998 for $R_{scc} = 250$ (balanced three phase equipment).				
^b Harmonic currents with a duration of up to 10 s maximum, in the case that the system is manually or automatically switched on for the first time or is definitely switched off.				

7 Documentation for the installer of the apparatus/assembly of apparatus

The installer shall be provided with documentary information and instructions for installation and use so that compliance with this standard is maintained.

This shall include where applicable:

- instructions for assembly and physical arrangement with other apparatus;
- instructions and precautions for interconnection to other apparatus;
- specifications of interconnection cables and devices, especially about the application of shielded cables;
- instructions for commissioning and testing;
- guidance on avoiding incorrect actions and assembly of apparatus which are known to cause non compliance with this standard.

Annex ZA **(informative)**

Relationship between this European Standard and the Essential Requirements of EU Directives

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide one means of conforming to Essential Requirements of the following EU Directive:

— EMC Directive 89/336/EEC, as amended by Directives 91/263/EEC, 92/31/EEC, 93/68/EEC and 93/97/EEC;

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the normative clauses of this standard confers, within the limits of the scope of this standard, a presumption of conformity with the relevant Essential Requirements of the EMC Directive and associated EFTA regulations.

WARNING — Other requirements and other EU Directives may be applicable to the product(s) falling within the scope of this standard.

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